



CSS

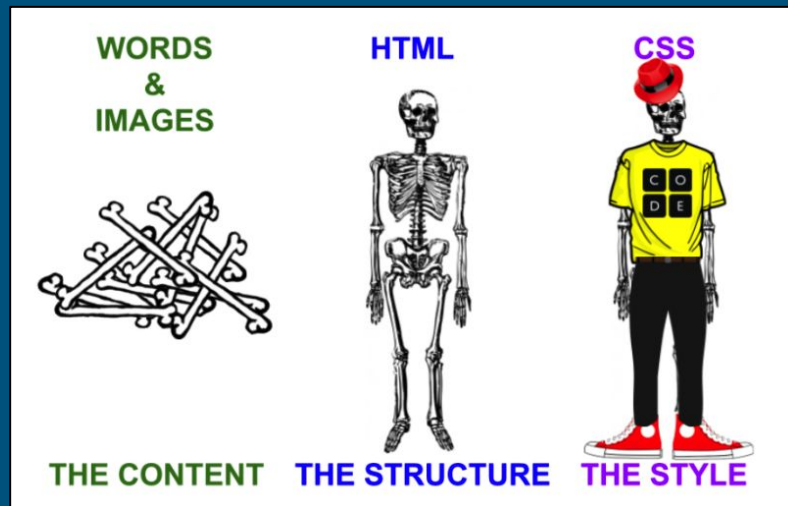


COM/CSC 271
Week 5: CSS Basics



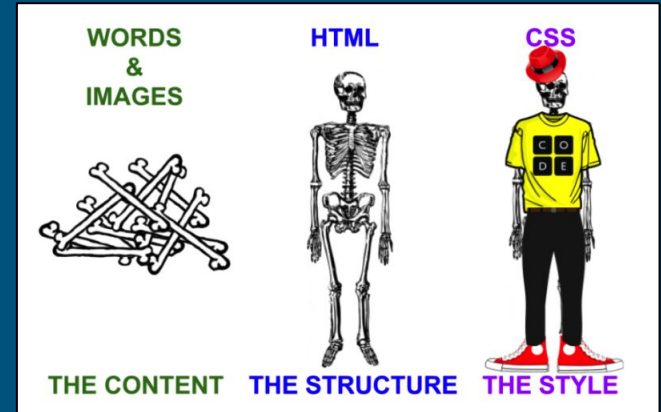
The Need For CSS

- So far, we have only made web pages where we control the content and structure.
- We have been using HTML as the language to specify the content and structure of the pages.
- While HTML allows us some control over how the page looks, it doesn't give developers much control over the specific look and style of each element.
- To do that, we need a language to express **style**—we need **CSS**.



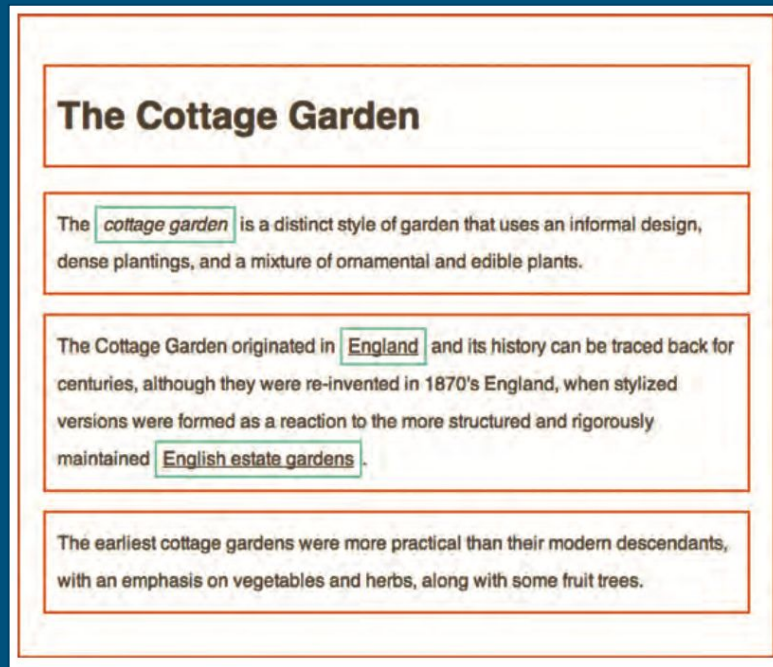
Why Use CSS?

- **Cascading Style Sheets (CSS)** is used to define **styles** for your web pages, including the design, layout and variations in display for different devices and screen sizes.
- HTML was NEVER intended to contain tags for formatting a web page! HTML was created to describe the content of a web page.
 - Adding font and color attributes to HTML 3.2 started a nightmare for web developers. Developing large websites with this information to each page became a long and expensive process.
- CSS removed the style formatting from the HTML page!



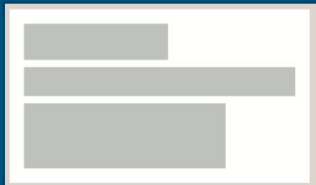
Understanding CSS: Thinking Inside The Box

- **CSS** is a language that adds **style** to a web page.
- The key to understanding how CSS works is to imagine that there is an **invisible box** around every HTML element.
- The HTML page has added outlines to each of the elements so that you can see how CSS will treat each element as if it lives inside its own box.



Block Elements

- Some elements will always appear to **start on a new line** in the browser window.
- These are known as **block level** elements.



- Examples of block elements are `<h1>`, `<p>`, `<ul>`, and `<li>`.

chapter-08/block-elements.html HTML

```
<h1>Hiroshi Sugimoto</h1>
<p>The dates for the ORIGIN OF ART exhibition are as follows:</p>
<ul>
  <li>Science: 21 Nov - 20 Feb 2010/11</li>
  <li>Architecture: 6 Mar - 15 May 2011</li>
  <li>History: 29 May - 21 Aug 2011</li>
  <li>Religion: 28 Aug - 6 Nov 2011</li>
</ul>
```

RESULT

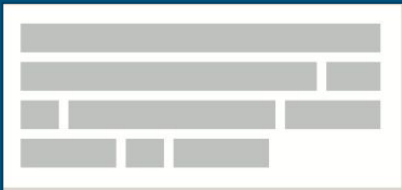
Hiroshi Sugimoto

The dates for the ORIGIN OF ART exhibition are as follows:

- Science: 21 Nov - 20 Feb 2010/11
- Architecture: 6 Mar - 15 May 2011
- History: 29 May - 21 Aug 2011
- Religion: 28 Aug - 6 Nov 2011

Inline Elements

- Some elements will always appear to **continue on the same line** as their neighboring elements.
- These are known as **inline** elements.



- Examples of inline elements are `<a>`, `<em>`, and `<img>`.

HTML chapter-08/inline-elements.html

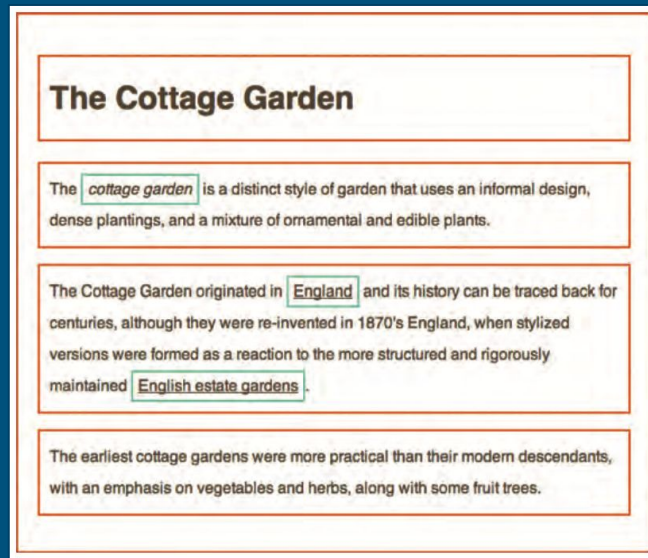
Timed to a single revolution of the planet around the sun at a 23.4 degrees tilt that plays out the rhythm of the seasons, this `<em>`Origins of Art`</em>` cycle is organized around four themes: `<b>`science, architecture, history`</b>` and `<b>`religion`</b>`.

RESULT

Timed to a single revolution of the planet around the sun at a 23.4 degrees tilt that plays out the rhythm of the seasons, this *Origins of Art* cycle is organized around four themes: **science, architecture, history and religion.**

Understanding CSS: Thinking Inside The Box

- **Block level** elements are shown with red borders.
- **Inline elements** have green borders.
- The `<body>` element creates the first box, then the `<h1>`, `<p>`, `<em>`, and `<a>` elements each create their own boxes within it.
- Using CSS, you could add a border around any of the boxes, specify its width and height, add a background color, and control text inside a box (i.e., color, size, typeface).



EXAMPLE STYLES

BOXES

Width and height
Borders (color, width, and style)
Background color and images
Position in the browser window.

TEXT

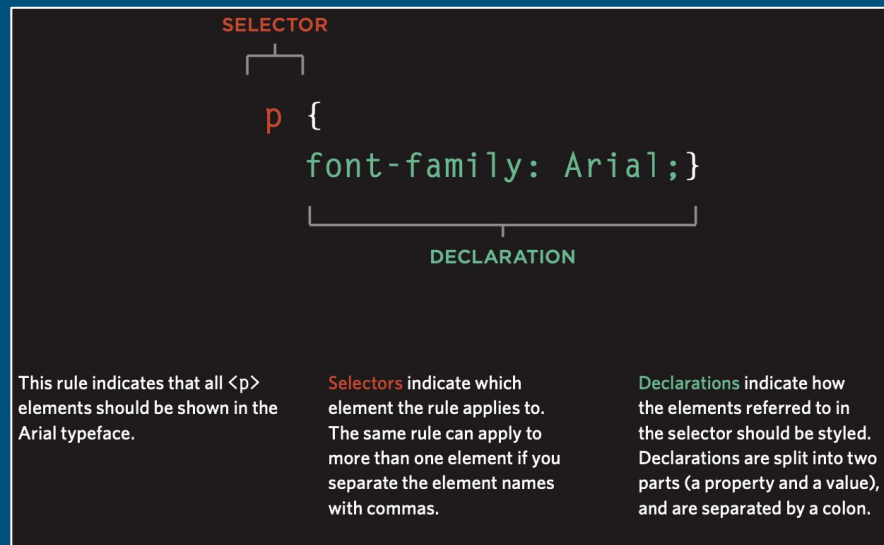
Typeface
Size
Color
Italics, bold, uppercase, lowercase, small-caps

SPECIFIC

There are also specific ways in which you can style certain elements such as lists, tables, and forms.

CSS Associates Style Rules With HTML Elements

- CSS works by associating **rules** with HTML elements.
- These **rules** govern **how** the **content** of specified elements should be **displayed**.
- A **CSS rule** contains two parts: a **selector** and a **declaration**.



CSS Properties Affect How Elements Are Displayed

- **CSS declarations** sit inside curly brackets and each is made up of two parts: a **property** and a **value**, separated by a colon.
- You can specify several properties in one declaration, each separated by a semi-colon.

```
h1, h2, h3 {  
    font-family: Arial;  
    color: yellow;  
}
```



The diagram shows a CSS declaration block. Below the code, there are two brackets. The first bracket is under 'font-family' and is labeled 'PROPERTY'. The second bracket is under 'yellow' and is labeled 'VALUE'.

This rule indicates that all `<h1>`, `<h2>` and `<h3>` elements should be shown in the Arial typeface, in a yellow color.

Properties indicate the aspects of the element you want to change. For example, color, font, width, height and border.

Values specify the settings you want to use for the chosen properties. For example, if you want to specify a color property then the value is the color you want the text in these elements to be.

CSS Example

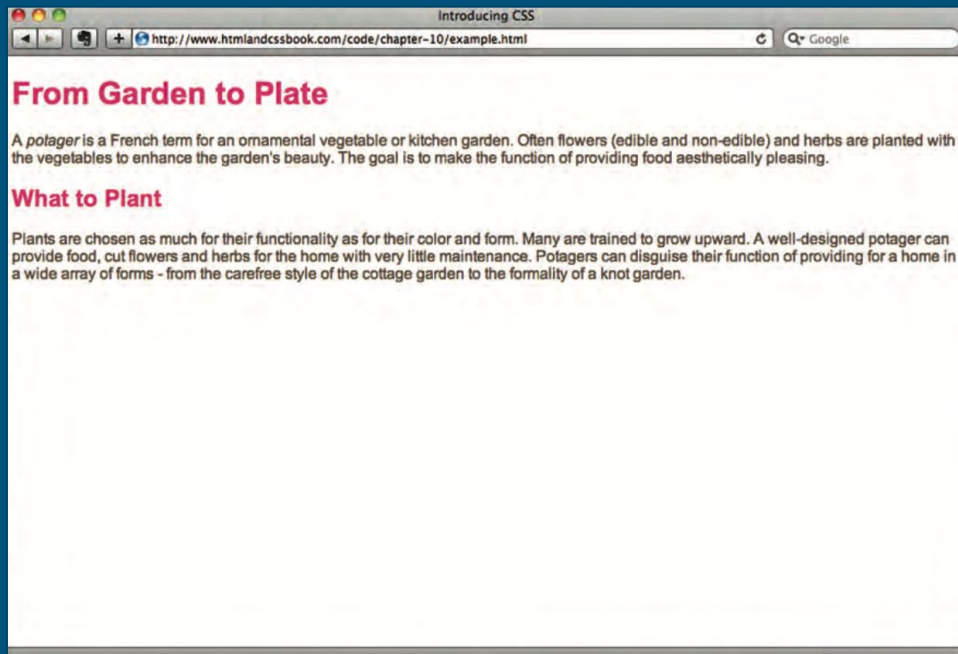
Here you can see a simple web page that is styled using CSS.

This example uses two documents: the HTML file (example.html) and a separate CSS file (example.css). The fifth line of HTML uses the <link> element to indicate where the CSS file is located.

On the next page, you will see how CSS rules can also be placed in your HTML pages and we will discuss when you might want to do this.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Introducing CSS</title>
    <link href="css/example.css" type="text/css"
          rel="stylesheet" />
  </head>
  <body>
    <h1>From Garden to Plate</h1>
    <p>A <i>potager</i> is a French term for an
      ornamental vegetable or kitchen garden ... </p>
    <h2>What to Plant</h2>
    <p>Plants are chosen as much for their functionality
      as for their color and form ... </p>
  </body>
</html>
```

```
body {
  font-family: Arial, Verdana, sans-serif;}
h1, h2 {
  color: #ee3e80;}
p {
  color: #665544;}
```



The CSS element Selector

- The **element selector** selects HTML elements based on the element name.
- **All <p> elements** on the page will be center-aligned with a red text color.

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<p>Every paragraph will be affected by the style.</p>
<p>Me too!</p>
<p>And me!</p>

</body>
</html>
```

Every paragraph will be affected by the style.

Me too!

And me!

The CSS id Selector

- The **id** of an element is **unique** within a page, so the **id selector** is used to select **one unique element!**
- To select an element with a **specific id**, write a **#** followed by the **id** of the element.
- The CSS rule will be applied to the **HTML element with id="para1"**.

<pre><!DOCTYPE html> <html> <head> <style> #para1 { text-align: center; color: red; } </style> </head> <body> <p id="para1">Hello World!</p> <p>This paragraph is not affected by the style.</p> </body> </html></pre>	Hello World! This paragraph is not affected by the style.
--	---

The CSS class Selector

- The **class selector** can select **multiple** HTML elements with a specific **class** attribute.
- To select elements with a specific **class**, write a **.** followed by the **class name**.
- **Only <p> elements with class="center"** will be red and center-aligned.

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned.</p>

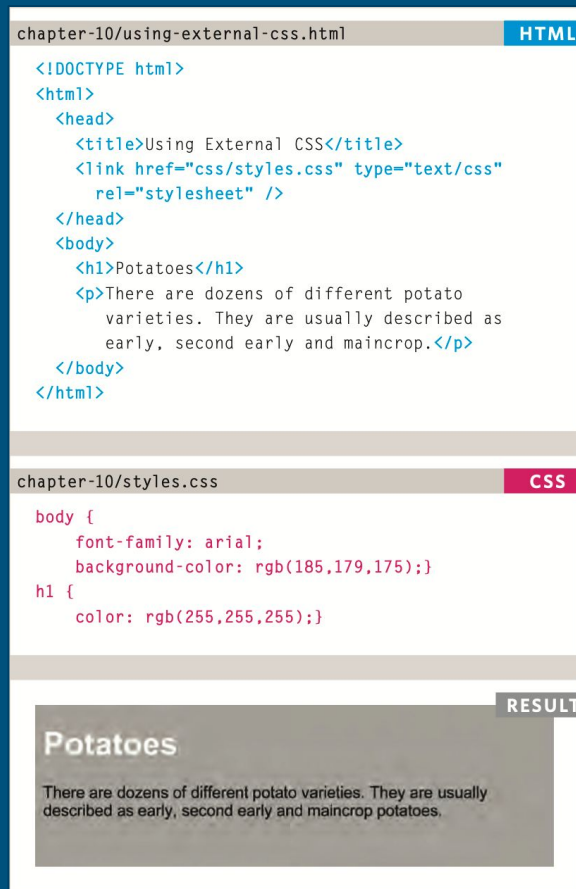
</body>
</html>
```

This heading will not be affected

This paragraph will be red and center-aligned.

Using External CSS

- The `<link>` element can be used in an HTML document to tell the browser **where to find the CSS file** used to style the page.
- It is an **empty element** (meaning it does not need a closing tag), and it lives **inside the `<head>`** element.
- It should use three attributes:
 - **href** - This specifies the **path to the CSS file** (which is often placed in a folder called **css** or **styles**).
 - **type** - This attribute specifies the **type of document** being linked to. The value should be **text/css**.
 - **rel** - This specifies the **relationship between the HTML page and the file** it is **linked to**. The value should be **stylesheet** when linking to a CSS file.



Using Internal CSS

- You can also **include CSS rules within an HTML page** by placing them inside a `<style>` element, which usually sits **inside the <head>** element of the page.
- The `<style>` element should use the **type** attribute to indicate that the styles are specified in CSS, and the value should be **text / css**.
- When building **a site with more than one page**, you should **use an external CSS style sheet**:
 - Allows all pages to use the same style rules (rather than repeating them in each page).
 - Keeps the content separate from how the page looks.
 - Means you can change the styles used across all pages by altering just one file (rather than each individual page).

